Problem 2: Cross-Entropy Is a Proper Scoring Rule

Claim. Let $Y \in \{0, 1\}$ and let the prediction $f(x) \in [0, 1]$ be the predicted probability that $Y = 1$. With the cross-entropy loss

$$L(y, q) = -\big[\, y \log q + (1 - y) \log(1 - q) \,\big],$$

the minimizer of the risk

$$f^* := \arg\min_f \; \mathbb{E}_{X,Y}\big[L(Y, f(X))\big]$$

is $f^*(x) = P(Y = 1 \mid X = x)$. (Throughout, we use the convention $0 \cdot \log 0 = 0$.)

By the tower property of expectation,

$$\mathbb{E}_{X,Y}\big[L(Y, f(X))\big] = \mathbb{E}_X\Big[\, \mathbb{E}\big[L(Y, f(X)) \mid X\big] \,\Big].$$

The outer expectation is an average over x. So if for *every fixed* x we choose the value $f(x)$ that minimizes the inner conditional expectation, the overall risk is minimized as well. This turns a minimization over all functions f into a minimization over a single number $q = f(x)$ for each x. (This is the same argument used in class to show that the minimizer of square error loss is $\mathbb{E}[Y \mid X]$.)

Fix x, and write

$$\eta := P(Y = 1 \mid X = x), \qquad q := f(x).$$

Given $X = x$, we have $Y = 1$ with probability η, in which case $L = -\log q$, and $Y = 0$ with probability $1 - \eta$, in which case $L = -\log(1 - q)$. Therefore

$$g(q) := \mathbb{E}\big[L(Y, q) \mid X = x\big] = -\eta \log q - (1 - \eta) \log(1 - q).$$

We need to show that g is minimized at $q = \eta$.

First suppose $0 < \eta < 1$. For $q \in (0, 1)$, using $\frac{d}{dq} \log q = \frac{1}{q}$ and, by the chain rule, $\frac{d}{dq} \log(1-q) = -\frac{1}{1-q}$,

$$g'(q) = -\frac{\eta}{q} + \frac{1 - \eta}{1 - q}.$$

Setting $g'(q) = 0$:

$$\begin{aligned} \frac{1 - \eta}{1 - q} &= \frac{\eta}{q} \\ (1 - \eta)\, q &= \eta\, (1 - q) \\ q - \eta q &= \eta - \eta q \\ q &= \eta. \end{aligned}$$

Since $0 < \eta < 1$, this stationary point lies inside $(0, 1)$.

Differentiating again (power rule plus chain rule; for the second term, both the exponent -1 and the inner derivative of $1-q$ contribute a factor of -1):

$$g''(q) = \frac{\eta}{q^2} + \frac{1-\eta}{(1-q)^2}.$$

For $q \in (0,1)$ both denominators are positive, and both numerators are nonnegative. They cannot both be zero, since $\eta + (1-\eta) = 1$. Hence

$$g''(q) > 0 \quad \text{for all } q \in (0,1),$$

so g is strictly convex on $(0,1)$, and its stationary point $q = \eta$ is the unique global minimizer on $(0,1)$. At the endpoints, $g(q) \to +\infty$ as $q \to 0$ or $q \to 1$ (because $\eta > 0$ and $1-\eta > 0$), so the endpoints are not better. Thus $q = \eta$ is the unique minimizer on $[0,1]$.

If $\eta = 1$, then $g(q) = -\log q$, which is strictly decreasing on $(0,1]$ and equals 0 (its smallest possible value, since $L \geq 0$) at $q = 1$. So the minimizer is $q = 1 = \eta$. Symmetrically, if $\eta = 0$, then $g(q) = -\log(1-q)$, which is minimized at $q = 0 = \eta$. So in every case the minimizer is $q = \eta$.

For every x, the inner conditional expected loss is minimized by setting $f(x) = \eta(x) = P(Y = 1 \mid X = x)$. Formally, for any predictor f,

$$\mathbb{E}\big[L(Y, f(X)) \mid X = x\big] \;\geq\; \mathbb{E}\big[L(Y, \eta(X)) \mid X = x\big] \quad \text{for every } x,$$

and taking the expectation over X on both sides gives $\mathbb{E}_{X,Y}[L(Y, f(X))] \geq \mathbb{E}_{X,Y}[L(Y, \eta(X))]$. Therefore

$$\boxed{f^*(x) = P(Y = 1 \mid X = x)}$$

and the cross-entropy loss is a proper scoring rule for binary Y. □